

Pacific Southwest Airlines Flight 182

Flight model: Boeing 727
Date: September 25, 1978

From: Sacramento, US
To: San Diego, US

2nd flight for the pilots. Sacramento to San Diego via Los Angeles. San Diego is a very difficult terrain and very busy airport. Flight 182 is assigned to land in runway 27.

Problem

The ATC informs the pilot that there is a Cessna flight and instructs the pilot for a "Visual separation", which means the pilots become responsible for the separation between the flights. The pilot turns left parallel to the runway. Later, the ATC radar shows that the Boeing 727 had collided with the Cessna. Finally crashing.

Investigation

The Voice recorder shows that pilots were aware of each other's position. The airport was fixed with a safety instrument that is designed to prevent such kind of incidents. The ATC says they ignored the warning which did pop up for both the planes saying the system was new and often had popped false alarm also, the pilots were informed regarding the inbound traffic. They also did contact the Cessna to inform the same. But, at exact moment they collided. The flight path of both the planes show that the flight 182 had collided with the Cessna. The pilots of the flight 182 assumed that the Cessna had passed to its right, which was never done. The flight path also shows that the Cessna turned right into the flight path of flight 182 just before the crash. Also, the Cessna student pilot was practicing control using instruments therefore was wearing a training hood, which covers the pilot's vision thereby helping him focus on the flight instruments. It was found that the Cessna was flying the same path as the flight 182. It was the job of the flight 182 crew to maintain the distance but as the plane such as Boeing 727 usually fly in a nose up position the Cessna was not visually visible to them. The investigators also plotted the view of the pilots of the flight 182 to understand if there were any blind spots. It seems that the Cessna should have been clearly visible to the crew of flight 182. It was found that all the pilots of flight 182 were busy in getting the landing checklist done and failed to visually spot the Cessna and maintain separation. It was also found that the pilot must have adjusted their seat as per their comfort and not as per the manufacturer's required position. It was studied that the varied seat position the Cessna was visible only for few seconds. And the terrain also made the Cessna camouflaged. The crew did try to contact the ATC but was not able to visually locate the Cessna on a casual note but, due to static noise there was a mis-communication between the two leading to a mis-interpretation. The flight 182 crew when they lost visual approach of separation of the Cessna, they must have informed the ATC which they had failed to do. The Cessna pilot had also changed the course without notifying ATC. The ATC should have warned the crew of flight 182 about the Cessna on its flight path.

Changes that were brought after the incident

The report advises to improve the Air traffic control system in the area. The airline also put on new rules on crew conversation when flying below 10,000 feet. Years after this incident, **TCAS (Traffic collision avoidance system)** was installed on all modern planes, which warns pilot when another plane comes too close.

Reference

1. How a Misheard Word Caused a Plane Crash — Mayday: Air Disaster
<https://www.youtube.com/watch?v=FK5SDp-Zx54>